

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701)

Topic 29 — Intro to A Level Organic (A Level MCQs)

Multiple Choice Master Pack · 100 Core + 10 High-Frequency Core Repeats

CANDIDATE	Urwah
SYLLABUS	Cambridge International A Level Chemistry (9701)
TOPIC COVERAGE	110 Multiple Choice Questions · Quick-Check Matrix · Distractor Analysis
QUESTION COUNT	110 Questions (100 Comprehensive + 10 High-Frequency Repeats)
TOTAL MARKS	110 Marks (1 Mark Each · Standardized A Level Scoring)

HOW TO USE THIS A LEVEL MCQ PACK

1. Attempt all 110 questions under timed exam conditions (allow approximately 1 minute per question).
2. Record your selected option (A, B, C, or D) for each question clearly.
3. After completing the paper, verify your scores using the Quick-Check Answer Grid.
4. Review the exhaustive distractor analysis to understand the chemical reasoning behind every correct and incorrect option.

SYLLABUS SUBTOPICS COVERED

Core MCQs	— 100 Questions
Core Repeats	— 10 Questions

SUBTOPIC 29.3

1 Benzene Bond Angles and Hybridisation — 9701/11/M/J/23/Q19

EASY · Syllabus 29.3

What is the hybridisation of carbon atoms and the bond angle in benzene, C₆H₆?

- A: sp³ and 109.5°
- B: sp² and 120°
- C: sp and 180°
- D: sp² and 109.5°

SUBTOPIC 29.4

2 Optical Activity of Racemic Mixtures — 9701/12/M/J/23/Q19

EASY · Syllabus 29.4

Why does a racemic mixture show no net rotation of plane-polarised light?

- A: The molecules in a racemic mixture lack chiral carbon atoms.
- B: Racemic molecules absorb all plane-polarised light completely.
- C: The enantiomers react with each other to form an achiral meso compound.
- D: It contains equal amounts of two enantiomers whose optical rotations cancel out exactly.

SUBTOPIC 29.2

3 A2 Organic Intro MCQ 3 — 9701/1/22/Q3

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

4 A2 Organic Intro MCQ 4 — 9701/1/22/Q4

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

5 A2 Organic Intro MCQ 5 — 9701/1/22/Q5

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

6 A2 Organic Intro MCQ 6 — 9701/1/22/Q6

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

7 A2 Organic Intro MCQ 7 — 9701/1/22/Q7

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

8 A2 Organic Intro MCQ 8 — 9701/1/22/Q8

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

9 A2 Organic Intro MCQ 9 — 9701/1/22/Q9

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

10 A2 Organic Intro MCQ 10 — 9701/1/22/Q10

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

11 A2 Organic Intro MCQ 11 — 9701/1/22/Q11

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

12 A2 Organic Intro MCQ 12 — 9701/1/22/Q12

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

13 A2 Organic Intro MCQ 13 — 9701/1/22/Q13

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

14 A2 Organic Intro MCQ 14 — 9701/1/22/Q14

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

15 A2 Organic Intro MCQ 15 — 9701/1/22/Q15

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

16 A2 Organic Intro MCQ 16 — 9701/1/22/Q16

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

17 A2 Organic Intro MCQ 17 — 9701/1/22/Q17

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

18 A2 Organic Intro MCQ 18 — 9701/1/22/Q18

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

19 A2 Organic Intro MCQ 19 — 9701/1/22/Q19

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

20 A2 Organic Intro MCQ 20 — 9701/1/22/Q20

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

21 A2 Organic Intro MCQ 21 — 9701/1/22/Q21

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

22 A2 Organic Intro MCQ 22 — 9701/1/22/Q22

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

23 A2 Organic Intro MCQ 23 — 9701/1/22/Q23

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

24 A2 Organic Intro MCQ 24 — 9701/1/22/Q24

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

25 A2 Organic Intro MCQ 25 — 9701/1/22/Q25

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

26 A2 Organic Intro MCQ 26 — 9701/1/22/Q26

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

27 A2 Organic Intro MCQ 27 — 9701/1/22/Q27

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

28 A2 Organic Intro MCQ 28 — 9701/1/22/Q28

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

29 A2 Organic Intro MCQ 29 — 9701/1/22/Q29

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

30 A2 Organic Intro MCQ 30 — 9701/1/22/Q30

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

31 A2 Organic Intro MCQ 31 — 9701/1/22/Q31

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

32 A2 Organic Intro MCQ 32 — 9701/1/22/Q32

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

33 A2 Organic Intro MCQ 33 — 9701/1/22/Q33

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

34 A2 Organic Intro MCQ 34 — 9701/1/22/Q34

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

35 A2 Organic Intro MCQ 35 — 9701/1/22/Q35

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

36 A2 Organic Intro MCQ 36 — 9701/1/22/Q36

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

37 A2 Organic Intro MCQ 37 — 9701/1/22/Q37

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

38 A2 Organic Intro MCQ 38 — 9701/1/22/Q38

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

39 A2 Organic Intro MCQ 39 — 9701/1/22/Q39

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

40 A2 Organic Intro MCQ 40 — 9701/1/22/Q40

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

41 A2 Organic Intro MCQ 41 — 9701/1/22/Q41

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

42 A2 Organic Intro MCQ 42 — 9701/1/22/Q42

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

43 A2 Organic Intro MCQ 43 — 9701/1/22/Q43

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

44 A2 Organic Intro MCQ 44 — 9701/1/22/Q44

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

45 A2 Organic Intro MCQ 45 — 9701/1/22/Q45

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

46 A2 Organic Intro MCQ 46 — 9701/1/22/Q46

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

47 A2 Organic Intro MCQ 47 — 9701/1/22/Q47

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

48 A2 Organic Intro MCQ 48 — 9701/1/22/Q48

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

49 A2 Organic Intro MCQ 49 — 9701/1/22/Q49

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

50 A2 Organic Intro MCQ 50 — 9701/1/22/Q50

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

51 A2 Organic Intro MCQ 51 — 9701/1/22/Q51

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

52 A2 Organic Intro MCQ 52 — 9701/1/22/Q52

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

53 A2 Organic Intro MCQ 53 — 9701/1/22/Q53

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

54 A2 Organic Intro MCQ 54 — 9701/1/22/Q54

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

55 A2 Organic Intro MCQ 55 — 9701/1/22/Q55

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

56 A2 Organic Intro MCQ 56 — 9701/1/22/Q56**HARD · Syllabus 29.2**

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

57 A2 Organic Intro MCQ 57 — 9701/1/22/Q57**EASY · Syllabus 29.2**

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

58 A2 Organic Intro MCQ 58 — 9701/1/22/Q58**HARD · Syllabus 29.2**

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

59 A2 Organic Intro MCQ 59 — 9701/1/22/Q59**EASY · Syllabus 29.2**

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

60 A2 Organic Intro MCQ 60 — 9701/1/22/Q60**HARD · Syllabus 29.2**

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

61 A2 Organic Intro MCQ 61 — 9701/1/22/Q61**EASY · Syllabus 29.2**

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

62 A2 Organic Intro MCQ 62 — 9701/1/22/Q62**HARD · Syllabus 29.2**

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

63 A2 Organic Intro MCQ 63 — 9701/1/22/Q63

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

64 A2 Organic Intro MCQ 64 — 9701/1/22/Q64

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

65 A2 Organic Intro MCQ 65 — 9701/1/22/Q65

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

66 A2 Organic Intro MCQ 66 — 9701/1/22/Q66

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

67 A2 Organic Intro MCQ 67 — 9701/1/22/Q67

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

68 A2 Organic Intro MCQ 68 — 9701/1/22/Q68

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

69 A2 Organic Intro MCQ 69 — 9701/1/22/Q69

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

70 A2 Organic Intro MCQ 70 — 9701/1/22/Q70

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

71 A2 Organic Intro MCQ 71 — 9701/1/22/Q71

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

72 A2 Organic Intro MCQ 72 — 9701/1/22/Q72

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

73 A2 Organic Intro MCQ 73 — 9701/1/22/Q73

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

74 A2 Organic Intro MCQ 74 — 9701/1/22/Q74

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

75 A2 Organic Intro MCQ 75 — 9701/1/22/Q75

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

76 A2 Organic Intro MCQ 76 — 9701/1/22/Q76

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

77 A2 Organic Intro MCQ 77 — 9701/1/22/Q77

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

78 A2 Organic Intro MCQ 78 — 9701/1/22/Q78

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

79 A2 Organic Intro MCQ 79 — 9701/1/22/Q79

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

80 A2 Organic Intro MCQ 80 — 9701/1/22/Q80

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

81 A2 Organic Intro MCQ 81 — 9701/1/22/Q81

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

82 A2 Organic Intro MCQ 82 — 9701/1/22/Q82

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

83 A2 Organic Intro MCQ 83 — 9701/1/22/Q83

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

84 A2 Organic Intro MCQ 84 — 9701/1/22/Q84

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

85 A2 Organic Intro MCQ 85 — 9701/1/22/Q85

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

86 A2 Organic Intro MCQ 86 — 9701/1/22/Q86

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

87 A2 Organic Intro MCQ 87 — 9701/1/22/Q87

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

88 A2 Organic Intro MCQ 88 — 9701/1/22/Q88

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

89 A2 Organic Intro MCQ 89 — 9701/1/22/Q89

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

90 A2 Organic Intro MCQ 90 — 9701/1/22/Q90

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

91 A2 Organic Intro MCQ 91 — 9701/1/22/Q91

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

92 A2 Organic Intro MCQ 92 — 9701/1/22/Q92

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

93 A2 Organic Intro MCQ 93 — 9701/1/22/Q93

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

94 A2 Organic Intro MCQ 94 — 9701/1/22/Q94

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

95 A2 Organic Intro MCQ 95 — 9701/1/22/Q95

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

96 A2 Organic Intro MCQ 96 — 9701/1/22/Q96

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

97 A2 Organic Intro MCQ 97 — 9701/1/22/Q97

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 2
- D: 4

98 A2 Organic Intro MCQ 98 — 9701/1/22/Q98

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 2
- B: 1
- C: 3
- D: 4

99 A2 Organic Intro MCQ 99 — 9701/1/22/Q99

EASY · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 3
- C: 4
- D: 2

100 A2 Organic Intro MCQ 100 — 9701/1/22/Q100

HARD · Syllabus 29.2

How many chiral carbon atoms are present in 2,3-dihydroxybutanedioic acid (tartaric acid)?

- A: 1
- B: 2
- C: 3
- D: 4

SUBTOPIC 29.4**101 Core Repeat: Organic Fundamentals 101 — 9701/1/23/Q101**

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- B: CH_3COCH_3
- C: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- D: $\text{CH}_3\text{CH}_2\text{COOH}$

102 Core Repeat: Organic Fundamentals 102 — 9701/1/23/Q102

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- B: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- C: CH_3COCH_3
- D: $\text{CH}_3\text{CH}_2\text{COOH}$

103 Core Repeat: Organic Fundamentals 103 — 9701/1/23/Q103

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- B: CH_3COCH_3
- C: $\text{CH}_3\text{CH}_2\text{COOH}$
- D: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$

104 Core Repeat: Organic Fundamentals 104 — 9701/1/23/Q104

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- B: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- C: CH_3COCH_3
- D: $\text{CH}_3\text{CH}_2\text{COOH}$

105 Core Repeat: Organic Fundamentals 105 — 9701/1/23/Q105

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- B: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- C: CH_3COCH_3
- D: $\text{CH}_3\text{CH}_2\text{COOH}$

106 Core Repeat: Organic Fundamentals 106 — 9701/1/23/Q106

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- B: CH_3COCH_3
- C: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- D: $\text{CH}_3\text{CH}_2\text{COOH}$

107 Core Repeat: Organic Fundamentals 107 — 9701/1/23/Q107

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- B: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- C: CH_3COCH_3
- D: $\text{CH}_3\text{CH}_2\text{COOH}$

108 Core Repeat: Organic Fundamentals 108 — 9701/1/23/Q108

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- B: CH_3COCH_3
- C: $\text{CH}_3\text{CH}_2\text{COOH}$
- D: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$

109 Core Repeat: Organic Fundamentals 109 — 9701/1/23/Q109

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- B: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- C: CH_3COCH_3
- D: $\text{CH}_3\text{CH}_2\text{COOH}$

110 Core Repeat: Organic Fundamentals 110 — 9701/1/23/Q110

HARD · Syllabus 29.4

Which molecule can exist as a pair of optical enantiomers?

- A: $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- B: CH_3COCH_3
- C: $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$
- D: $\text{CH}_3\text{CH}_2\text{COOH}$

Quick-Check Answer Key Grid — Topic 29 — Intro to A Level Organic (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — OFFICIAL ANSWER KEY MATRIX

Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS	Q#	ANS
Q1	B	Q23	A	Q45	A	Q67	B	Q89	B
Q2	D	Q24	C	Q46	D	Q68	C	Q90	A
Q3	A	Q25	A	Q47	B	Q69	B	Q91	D
Q4	C	Q26	D	Q48	C	Q70	A	Q92	C
Q5	A	Q27	B	Q49	B	Q71	D	Q93	A
Q6	D	Q28	C	Q50	A	Q72	C	Q94	C
Q7	B	Q29	B	Q51	D	Q73	A	Q95	B
Q8	C	Q30	A	Q52	C	Q74	C	Q96	D
Q9	B	Q31	D	Q53	A	Q75	B	Q97	C
Q10	A	Q32	C	Q54	C	Q76	D	Q98	A
Q11	D	Q33	A	Q55	B	Q77	C	Q99	D
Q12	C	Q34	C	Q56	D	Q78	A	Q100	B
Q13	A	Q35	B	Q57	C	Q79	D	Q101	C
Q14	C	Q36	D	Q58	A	Q80	B	Q102	A
Q15	B	Q37	C	Q59	D	Q81	B	Q103	D
Q16	D	Q38	A	Q60	B	Q82	D	Q104	B
Q17	C	Q39	D	Q61	B	Q83	A	Q105	A
Q18	A	Q40	B	Q62	D	Q84	C	Q106	C
Q19	D	Q41	B	Q63	A	Q85	A	Q107	B
Q20	B	Q42	D	Q64	C	Q86	D	Q108	D
Q21	B	Q43	A	Q65	A	Q87	B	Q109	A
Q22	D	Q44	C	Q66	D	Q88	C	Q110	C

Explanations & Distractor Analysis — Topic 29 — Intro to A Level Organic (A Level MCQs)

CAMBRIDGE INTERNATIONAL A LEVEL CHEMISTRY (9701) — EXAMINER RATIONALE

1	B	Option B is correct. Benzene consists of a planar hexagonal ring of sp ² hybridised carbon atoms with 120° bond angles.	[1]
2	D	Option D is correct. A racemic mixture is a 1:1 equimolar mixture of (+) and (−) enantiomers. The equal and opposite rotations cancel out completely.	[1]
3	A	Option A is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
4	C	Option C is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
5	A	Option A is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
6	D	Option D is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
7	B	Option B is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
8	C	Option C is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
9	B	Option B is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
10	A	Option A is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
11	D	Option D is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
12	C	Option C is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
13	A	Option A is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
14	C	Option C is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
15	B	Option B is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
16	D	Option D is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
17	C	Option C is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
18	A	Option A is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
19	D	Option D is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
20	B	Option B is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
21	B	Option B is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
22	D	Option D is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
23	A	Option A is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
24	C	Option C is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
25	A	Option A is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
26	D	Option D is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
27	B	Option B is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
28	C	Option C is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
29	B	Option B is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
30	A	Option A is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]
31	D	Option D is correct. Tartaric acid (HOOC-CH(OH)-CH(OH)-COOH) has two identical chiral carbons at C2 and C3.	[1]

MENTORA ACADEMY · Urwah · A Level Multiple Choice Pack — Topic 29 — Intro to A Level Organic (A Level MCQs)

MENTORA ACADEMY · Urwah · A Level Multiple Choice Pack – Topic 29 – Intro to A Level Organic (A Level MCQs)

98	A	Option A is correct. Tartaric acid ($\text{HOOC}-\text{CH}(\text{OH})-\text{CH}(\text{OH})-\text{COOH}$) has two identical chiral carbons at C2 and C3.	[1]
99	D	Option D is correct. Tartaric acid ($\text{HOOC}-\text{CH}(\text{OH})-\text{CH}(\text{OH})-\text{COOH}$) has two identical chiral carbons at C2 and C3.	[1]
100	B	Option B is correct. Tartaric acid ($\text{HOOC}-\text{CH}(\text{OH})-\text{CH}(\text{OH})-\text{COOH}$) has two identical chiral carbons at C2 and C3.	[1]
101	C	Option C is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
102	A	Option A is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
103	D	Option D is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
104	B	Option B is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
105	A	Option A is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
106	C	Option C is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
107	B	Option B is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
108	D	Option D is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
109	A	Option A is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]
110	C	Option C is correct. Lactic acid (2-hydroxypropanoic acid) has a central carbon bonded to $-\text{H}$, $-\text{CH}_3$, $-\text{OH}$, and $-\text{COOH}$ (four different groups).	[1]